

Ethan Joseph

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EDUCATION

Bachelor of Engineering (B. Eng) – Software Engineering

McMaster University

Graduation Date: June 2030

WORK EXPERIENCE

Software Engineer Intern - DoneMaker

May 2026 – August 2026

- Owned and shipped **full-stack production features** in an existing **TypeScript, React, Node.js, and Supabase/PostgreSQL** codebase, taking requirements through implementation, testing, review, and deployment for a production workflow spanning **~160 blog records**.
- Designed the **backend architecture** for an authenticated, on-demand reporting workflow, reducing third-party API calls by **~65%** and cutting repeat report load times from **2.4s to 350ms** through **24-hour caching, request deduplication, retry/backoff, graceful failure handling, and secure secret management**.
- Built **automated unit and integration tests** covering **6+ critical workflow and failure paths**, including authentication, report generation, API aggregation, caching, retry behaviour, upstream failures, and regression cases.
- Containerized the application and test environment with **Docker** and integrated it into **GitHub Actions CI**, automatically running linting, tests, production builds, and integration checks on every pull request in **~4 minutes**.
- Worked through a professional **Git / pull-request workflow** with a **3-person development team**, using feature branches, syncing with main, resolving integration issues, participating in review/testing, and shipping the completed feature to production.

Founder & Organizer - Eclipse Hackathons - [View Devpost](#)

August 2024 – November 2024

- Founded and led an **80+ participant hackathon across 15+ Ontario schools**, securing HCLTech sponsorship and recruiting judges from AMD, Dropbox, Fractal Analytics, and SAP Concur.
- Designed and developed the event's website, registration system, submission portal, automated email workflows, and backend application infrastructure supporting **80+ registered users with zero downtime during live judging**.

PROJECTS

Sentinel Security Auditor – [View Live Site](#) | Python, TypeScript, Gemini API, GitHub Actions, Vercel

Nov 2025 – Present

- Built an AI-driven application security auditor that analyzes local and GitHub repositories to generate structured vulnerability findings, attack-flow graphs, and remediation guidance.
- Deployed a production web interface on Vercel, averaging **~800 monthly visits** from users scanning their repositories for security vulnerabilities.
- Developed a pipeline to analyze large codebases (**50k+ LOC**) by using prompt compression techniques, reducing token spend by 28%.
- Developed repository-aware vulnerability analysis that **reduced false positives by ~35%** vs rule-only scanners (benchmarked across 40+ open-source repos).
- Implemented automated remediation with structured Git patch generation, achieving **~70% successful fix** application rate as measured by post-scan verification and rescanning of patched code.

SafeGuard - [MacHacks Winner: Best Use of Gemini API](#) | JavaScript, React, Tailwind CSS, Flask, Gemini API, Vercel

Mar 2026

- Built and deployed an AI safety application that analyzes environmental images to identify hazards and generate structured risk reports.
- Engineered a Gemini vision pipeline with normalized JSON outputs supporting **up to 8 images per request** and context-aware follow-up analysis.
- Implemented client-side image compression and serverless API processing, **reducing average payload size by ~60%** before inference.

TECHNICAL SKILLS

Languages | Python, TypeScript/JavaScript, Java, SQL, C/C++

Frameworks & Technologies | React, Node.js, Flask, PostgreSQL, Supabase, REST APIs

AI Engineering | Gemini API, LLM Application Development, Structured Outputs, Prompt & Context Optimization, Multi-API Data Integration

Systems & Developer Tools | Git, GitHub Actions, Docker, Linux, Vercel, CI/CD, Caching, Rate-Limit Handling, Unit Testing, Debugging